



FileEditSelectionViewGoRunTerminalHelppython_test

Update

EXPLORER

PYTHON_TEST

DAY-04 Program Outp...

DAY-05 Program Outp...

DAY-06 Program Outp...

DAY-07 Program Outp...

DAY-08 Program Outp...

DAY-09 Program and ...

DAY-10 Program and ...

DAY-11 Program and ...

DAY-12 Program and ...

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exercise_2_color_picke...

exercise_2_compariso...

OUTLINE

TIMELINE

Welcome

exercise_1_contour_analysis.py

exercise_1_contour_analysis.py > ...

```
1 import cv2
2 import numpy as np
3
4 def analyze_contours(image_path):
5     """Analyze contours and print statistics"""
6     image = cv2.imread(image_path)
7     if image is None:
8         print(f"Error: Could not load image from {image_path}")
9         return None, [], []
10
11     gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
12     blurred = cv2.GaussianBlur(gray, (5, 5), 0)
13     _, binary = cv2.threshold(blurred, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU)
14
15     # FIX 1: Modern OpenCV returns (contours, hierarchy). No slicing needed.
16     contours, _ = cv2.findContours(binary, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
17
18     print("Contour Analysis Report")
19     print("-" * 50)
20
21     areas = []
```

PROBLEMS

OUTPUT

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```
PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise_1_contour_analysis.py
Contour Analysis Report
-----
Contour 1: Area=9082, Perimeter=409.8
Contour 2: Area=9082, Perimeter=409.8
Contour 3: Area=9660, Perimeter=471.7
Contour 4: Area=9660, Perimeter=471.7
-----
Total contours: 4
Total area: 37482 pixels
Average area: 9370
Largest area: 9660
Smallest area: 9082
PS C:\Users\raghu\OneDrive\Desktop\python_test>
```

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powercat...

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Python

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powershell

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UTF-8

CRLF

Python

3.14.5

Python 3.14 (64-bit)

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Update

EXPLORER

WELCOME

exercise_2_bounding_boxes.py X

PYTHON TEST

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- DAY-04 Program Outp...
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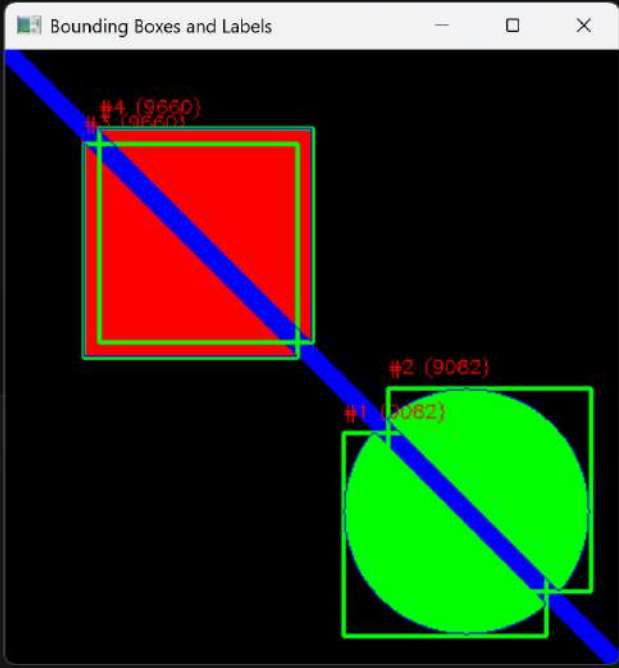
exercise_2_bounding_boxes.py > ...

```
4 def draw_bboxes_and_labels(image_path):
12     blurred = cv2.GaussianBlur(gray, (5, 5), 0)
13     _, binary = cv2.threshold(blurred, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU)
14
15     # FIX 1: Correct unpack for modern OpenCV
16     contours, _ = cv2.findContours(binary, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
17
18     result = image.copy()
19
20     for i, contour in enumerate(contours):
21         # Get bounding box
22         x, y, w, h = cv2.boundingRect(contour)
23
24         # Draw green bounding rectangle
25         cv2.rectangle(result, (x, y), (x+w, y+h), (0, 255, 0), 2)
26
27         # Draw blue contour inside it
28         cv2.drawContours(result, [contour], -1, (255, 0, 0), 1)
29
30         # Calculate area and format label
31         area = cv2.contourArea(contour)
32         label = f"#{i+1} ({area:.0f})"
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise_2_bounding_boxes.py

Bounding Boxes and Labels



python

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EXPLORER

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exercise1.brightness.c...
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exercise1.house.py
exercise1.py
exercise1.thumbnail.py
exercise2.day3.py
avarrice2.dissolve.nv

OUTLINE
TIMELINE

Welcome

exercise_3_crop_objects.py X

exercise_3_crop_objects.py > ...

5 def crop_and_save_objects(image_path, output_dir="cropped_objects"):
16 gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
17 blurred = cv2.GaussianBlur(gray, (5, 5), 0)
18 _, binary = cv2.threshold(blurred, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU)
19
20 # FIX 1: Correct unpack for modern OpenCV
21 contours, _ = cv2.findContours(binary, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
22
23 print(f"Cropping {len(contours)} objects...")
24
25 # FIX 2: Properly indented loop body so every object is processed
26 for i, contour in enumerate(contours):
27 x, y, w, h = cv2.boundingRect(contour)
28
29 # Add small padding
30 pad = 5
31 x_start = max(0, x - pad)
32 y_start = max(0, y - pad)
33 x_end = min(image.shape[1], x + w + pad)
34 y_end = min(image.shape[0], y + h + pad)

PROBLEMS

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PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise_3_crop_objects.py
Cropping 4 objects...
Saved: cropped_objects/object_1.png
Saved: cropped_objects/object_2.png
Saved: cropped_objects/object_3.png
Saved: cropped_objects/object_4.png
All objects saved to cropped_objects/
PS C:\Users\raghu\OneDrive\Desktop\python_test>

powerst...
Python
powersh...
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Ln 48, Col 24 Spaces: 4 UTF-8 CRLF Python 3,14.5 Python 3.14 (64-bit)

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WELCOME

exercise_4_shape_classifier.py

```
def classify_shapes(image_path):
    # Approximate contour
    perimeter = cv2.arcLength(contour, True)
    approx = cv2.approxPolyDP(contour, 0.04 * perimeter, True) # 0.04 is often a safer baseline
    vertices = len(approx)

    # Determine shape based on vertex count
    if vertices == 3:
        shape = "Triangle"
    elif vertices == 4:
        x, y, w, h = cv2.boundingRect(approx)
        aspect_ratio = w / float(h) if h > 0 else 1
        # A square has a roughly equal width and height
        if 0.95 <= aspect_ratio <= 1.05:
            shape = "Square"
        else:
            shape = "Rectangle"
    elif vertices == 5:
        shape = "Pentagon"
    elif vertices == 6:
        shape = "Hexagon"
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

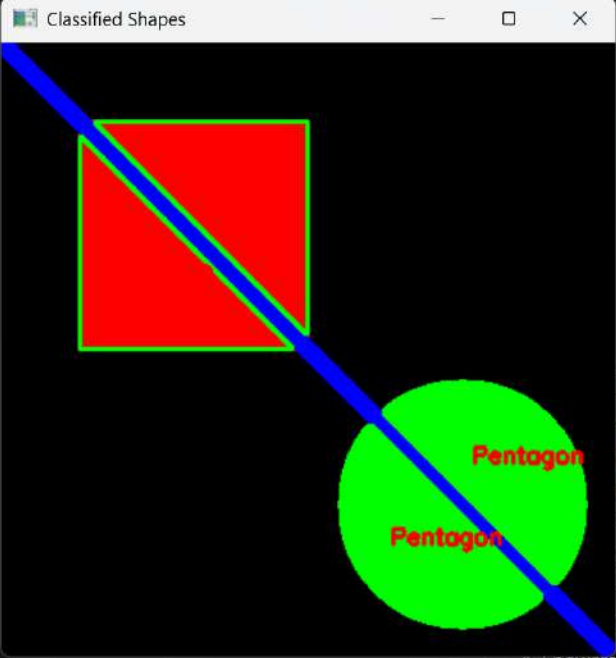
PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise_4_shape_classifier.py

Shape Classification Results:

Pentagon: 2

Triangle: 2

Classified Shapes



python

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exercise_4_histogram...

exercise_4_shape_clas...

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exercise_5_deblurring...

exercise_5_document...

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exercise1.day3.py

exercise1.house.py

exercise1.nv

OUTLINE

TIMELINE

Welcome

exercise_5_document_scanner.py X

exercise_5_document_scanner.py > ...

22 def scan_document(image_path):

87 # Warp the perspective

88 M = cv2.getPerspectiveTransform(ordered_pts, dst)

89 warped = cv2.warpPerspective(image, M, (width, height))

90

91 cv2.imshow("Scanned Document", warped)

92 cv2.imwrite("scanned_document.png", warped)

93 print("Scanned document saved successfully as scanned_document.png")

94 else:

95 print("Could not isolate a clean 4-cornered document outline. Check threshold/edges.")

96

97 cv2.waitKey(0)

98 cv2.destroyAllWindows()

99

100 # Run document scanner

101 scan_document("test_image.png")

PROBLEMS

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PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise_5_document_scanner.py

Document corners detected: 9

Could not isolate a clean 4-cornered document outline. Check threshold/edges.

Document Detected

python

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